ASSIGNMENT 1 – TASK 4: Design Classes, Objects and Constructors

Bloom's Level: K4 – Analyse | Marks: 4

A constructor initialises a new object. A default constructor takes no arguments; a parameterised constructor receives required data; an overloaded constructor offers multiple valid creation options. Java supplies a default constructor only when no constructor is written.

```
class ParkingTicket {
    private final long ticketNo;
    private final String vehicleNo;

    ParkingTicket(long ticketNo, String vehicleNo) {
        this.ticketNo = ticketNo;
        this.vehicleNo = vehicleNo;
    }

    ParkingTicket(ParkingTicket oldTicket, long newTicketNo) {
        this(newTicketNo, oldTicket.vehicleNo); // chain to main constructor
    }

    public long getTicketNo() { return ticketNo; }
    public String getVehicleNo() { return vehicleNo; }
}
```

this refers to the current object and this(…) calls another constructor in the same class. Constructor chaining avoids repeated initialisation. super(…) calls a superclass constructor and must be the first constructor statement. A copy constructor is not built into Java; use one only when copying makes sense. A reissued ticket should normally receive a new unique ticket number rather than duplicate the old identity.

Reference variables are local variables/fields that hold object references. Local reference variables typically live in stack frames; objects are allocated on the heap. Passing an object to a method passes its reference value. Creating n tickets uses O(n) time and O(n) heap space for the n objects.